

PHYS 143 – Mechanics and Thermal Physics

Vectors

How do we describe, combine and calculate with quantities that have both magnitude and direction?

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- **Text:** Young and Freedman, *University Physics* (15th ed.). Copies in the Physics Department Library and Balme Library.
- **Sakai** (sakai.ug.edu.gh): slides, problem sheets, announcements, office hours. Log in this week.
- **Attendance** is required; ten absences without a UG physician's note bars you from the final exam.
- **Homework** must be your own work. Asking lecturers, TAs and classmates for help is encouraged.

Everything else

Venues, times, TA names and the assessment breakdown are on Sakai. Extra material, practice questions and the Reference deck for each lecture: hhdonglo.github.io.

This week: the central question and the thread

Central question

How do we describe, combine and calculate with quantities that have both magnitude *and* direction?

By the end of this week, you should be able to:

1. Represent, add and subtract vectors graphically.
2. Resolve vectors into components and reconstruct their magnitude and direction.
3. Add vectors using components and unit-vector notation, and find the unit vector of any vector.
4. Calculate and interpret dot and cross products.

Reading: Young and Freedman, Chapter 1, §1.7–1.10. **Slide labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

The thread of the week

Vectors give us a way to represent quantities with magnitude and direction. **Graphs build understanding; components give us a systematic way to calculate.**

- 1 UNDERSTAND:** why do we need vectors?
2 CALCULATE: how do we calculate accurately?
3 EXTEND: what can two vectors construct?

An opening problem

Opening problem

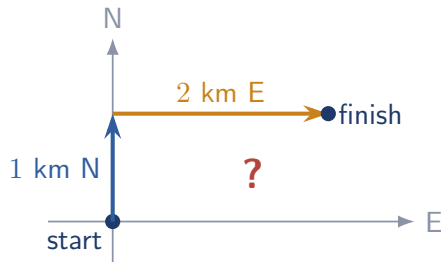
You walk **1 km north**, then **2 km east**.

Where are you now, relative to where you started?

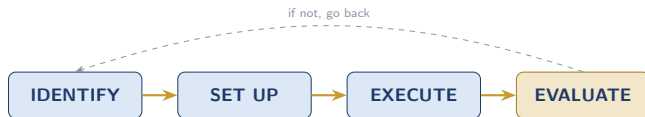
Predict first

- Are you 3 km from the starting point?
- Is a single number a complete answer?
- In approximately what direction should the answer point?

But what if... we cannot yet solve this completely. Before solving it, here is the method we will use throughout physics: **IDENTIFY, SET UP, EXECUTE, EVALUATE.**



We solve this once we can add vectors at 90° .



IDENTIFY

What is being asked?
What quantities are known?
What physics is relevant?

SET UP

Draw a diagram.
Choose coordinates.
Predict the approximate size and direction.
Select the equations.

EXECUTE

Perform the calculation.
Carry units.
Keep track of signs.

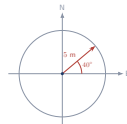
EVALUATE

Are the units correct?
Is the magnitude reasonable?
Does the direction agree with the sketch?
Does it make physical sense?

But what if... the method needs quantities to work with. What do “how much” and “which way” mean exactly? → **magnitude and direction.**

Contact 1: Understand

Why do we need vectors?



Magnitude: how much?

The **size** of a quantity: a number with units, such as 5 m, 70 kg, 15 m/s. Never negative. A bag is 5 m away: which marked point is it?

Direction: which way?

The **orientation** of a quantity relative to a reference: a compass word, or an angle from a reference axis.

Rule: a complete answer to “where?”

magnitude *and* direction: 5 m at 40° north of east.

But what if... do all quantities need a direction? Mass is 70 kg whichever way you face. Some need both, some only a size: **scalars and vectors**.

Scalar

A scalar quantity is **described completely by a numerical value and appropriate units**, where applicable.

e.g. mass, time, temperature, distance, speed, energy.

Notation: vector and magnitude

$\vec{A}$ = the vector

$A = |\vec{A}|$ = magnitude of the vector

In $\vec{A} \cdot \vec{B} = AB \cos \phi$, A and B are magnitudes.
 A_x is a number; $A_x \hat{i}$ is a vector.

Vector

A vector quantity **requires both magnitude and direction**.

e.g. displacement, velocity, acceleration, force.

Keep these distinct

The magnitude of a vector is a **scalar** and is **never negative**.

Decide for each, then compare with your neighbour

- | | |
|---------------------------|-------------------------------|
| (a) 15 m/s | (b) 15 m/s north |
| (c) distance walked, 3 km | (d) displacement, 3 km east |
| (e) temperature, 30°C | (f) a force of 20 N downwards |
| (g) -5 J of energy | (h) mass, 70 kg |

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Answers

Scalars: (a) speed, (c) distance, (e), (g), (h). Vectors: (b) velocity, (d) displacement, (f) force. **A negative number is not automatically a vector:** -5 J is a scalar.

Decide for each, then compare with your neighbour

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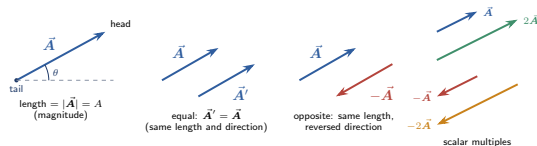
Answers

Scalars: (a) speed, (c) distance, (e), (g), (h). Vectors: (b) velocity, (d) displacement, (f) force. **A negative number is not automatically a vector:** -5 J is a scalar.

Distance and displacement

You walk once round a 400 m track and stop where you started. **Distance** (scalar) is the total path length: 400 m. **Displacement** (vector) runs directly from the initial to the final position: here **zero**.

But what if... how do we name, draw and compare vectors? → **representing a vector**.



Rule: equal and opposite vectors

Two vectors are **equal** if they have the same magnitude and the same direction, wherever they are drawn.

$-\vec{A}$ has the same magnitude as $\vec{A}$ and the opposite direction.

Rule: multiplying by a scalar

$c\vec{A}$ has magnitude $|c|A$.

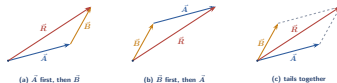
$c > 0$: the direction is unchanged ($2\vec{A}$).

$c < 0$: the direction is reversed ($-\vec{A}$, $-2\vec{A}$).

But what if... two displacements follow one another, as in the opening problem. What single displacement results? → **graphical addition**.

Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?



Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?



Rule: head to tail, (a) and (b)

Put the tail of the second vector at the head of the first. $\vec{R}$ runs from the first tail to the last head.

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \text{ (commutative);}$$

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C}) \text{ (associative).}$$

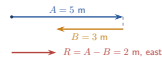
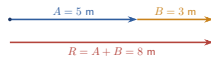
Rule: parallelogram, (c)

Draw both tails at one point. $\vec{R}$ is the *diagonal* from that point of the parallelogram built on $\vec{A}$ and $\vec{B}$.

But what if... how big is $\vec{R}$? Start with the simplest cases, vectors along one line.

Predict first: sketch 5 m east then 3 m east, and 5 m east then 3 m west. Predict each resultant.

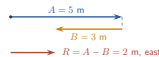
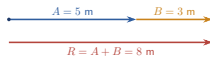
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Rules: parallel, antiparallel, and the bounds on R

Parallel: $|\vec{R}| = A + B$, the largest. **Antiparallel:** $|\vec{R}| = |A - B|$, along the *larger* vector, the smallest. In general $|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$: for 3 m and 4 m, $1 \text{ m} \leq R \leq 7 \text{ m}$. **Magnitudes do not simply add.**

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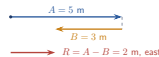
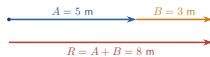
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Your turn

A hiker walks 4.5 km east, then (i) 1.5 km east, (ii) 6.0 km west. Predict, then find $\vec{R}$.

But what if... we want to undo an addition? $\rightarrow$ **subtraction.**

Predict first: sketch 5 m east then 3 m east, and 5 m east then 3 m west. Predict each resultant.



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Your turn

A hiker walks 4.5 km east, then (i) 1.5 km east, (ii) 6.0 km west. Predict, then find $\vec{R}$. (i) 6.0 km east; (ii) 1.5 km west

But what if... we want to undo an addition? $\rightarrow$ **subtraction.**

Rule: subtraction

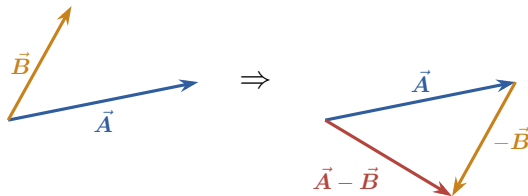
$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

Reverse $\vec{B}$, then add head to tail.

Meaning: $\vec{A} - \vec{B}$ takes you from the head of $\vec{B}$ to the head of $\vec{A}$ (shared tail).

Connect: Lecture 2

Change in velocity $\Delta \vec{v} = \vec{v}_2 - \vec{v}_1$, hence acceleration.



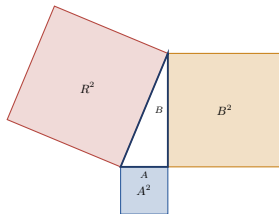
But what if... we need numbers, not drawings? A drawing is only as accurate as the ruler. Calculate, starting at 90° .

Predict first

You walk 5.0 m east, then 12 m north. Is the displacement 17 m? Predict its size and direction.

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Head to tail, the legs meet at 90° : a **right triangle**.

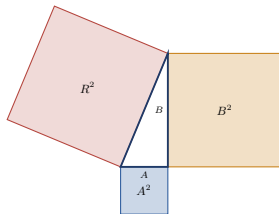
Prediction: $12 < R < 17$ m, leaning north.

Rule: perpendicular vectors

$$R = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1}(B/A), \quad \theta \text{ between } \vec{R} \text{ and } \vec{A}$$

Predict first

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$$R = \sqrt{5.0^2 + 12^2} = 13 \text{ m}, \quad \theta = \tan^{-1}(12/5.0) = 67^\circ \text{ north of east. } \checkmark (12 < 13 < 17)$$

But what if... does it work when the legs are not at 90° ? First, solve the opening problem.

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Execute:

$$R = \sqrt{(1.00 \text{ km})^2 + (2.00 \text{ km})^2} = 2.24 \text{ km}$$

$$\tan \theta = \frac{2.00 \text{ km}}{1.00 \text{ km}} \Rightarrow \theta = 63.4^\circ \text{ east of north}$$

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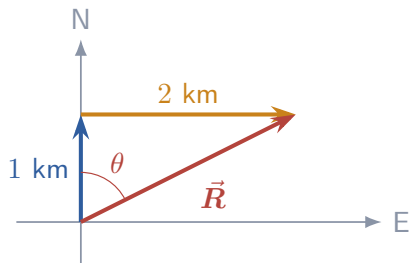
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Evaluate: units km; $2 < R < 3$ km as predicted; direction closer to east, matching the sketch.



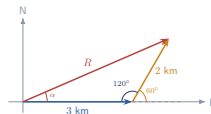
Young, Example 1.5

But what if... change the second leg to 2.00 km at 60° north of east, after 3.00 km east? Pythagoras would give $\sqrt{3.00^2 + 2.00^2} = 3.61$ km. Is that right?

Example 2: a cyclist

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.

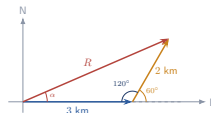
Predict first: is Pythagoras allowed? Is R nearer 5 or 3 km?



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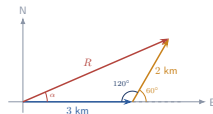


No right angle: the *internal* angle between the legs is $\gamma = 180^\circ - 60^\circ = 120^\circ$. Pythagoras would give 3.61 km, but the second leg alone carries the cyclist $2.00 \cos 60^\circ = 1.00$ km further east, so $R \geq 4.00$ km: 3.61 km is impossible.

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Rule: cosine rule

$$R^2 = A^2 + B^2 - 2AB \cos \gamma$$

$\gamma = 120^\circ$ is the internal angle of the head-to-tail triangle; the tail-to-tail angle between the legs is $\phi = 60^\circ$, used later in dot and cross products.

Execute:

$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00) \cos 120^\circ = 19.0$, so $R = 4.36$ km. **REFERENCE, optional route:** the sine rule gives the direction, $\sin \alpha = \frac{2.00 \sin 120^\circ}{4.36}$, so $\alpha = 23.4^\circ$ north of east. **Evaluate:** $3 < R < 5$. ✓

Question

A delivery rider makes four legs: 2.1 km at 20° N of E, 3.4 km at 35° W of N, 1.8 km S, 2.6 km at 10° S of W. Where does the rider end up?

- Graphical methods give physical understanding ...
- ... but scale drawings have limited precision,
- and each extra leg needs another triangle, with its own cosine and sine rule.

The transition to Contact 2

Triangle methods work, but become cumbersome. Components give a systematic numerical method for any number of vectors.

Common mistakes and the correct habit

Mistake**Instead**

Giving the magnitude a sign: $|\vec{A}| < 0$.

The magnitude is never negative; the minus sign belongs to the direction.

Adding magnitudes: $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$.

Add head to tail; then $|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$.

Calling any negative number a vector (-5 J).

Ask whether a direction is needed: energy is a scalar.

Using Pythagoras when the angle is not 90° .

Find the internal angle, then the cosine and sine rules.

Reading “ 30° north of east” from north.

Start from the first named direction and turn toward the second.

Confusing distance with displacement.

Displacement is final minus initial; distance is the path length.

Contact 2: Calculate

How do we calculate with vectors accurately?

Where we are: we can represent vectors and add them graphically and with triangles. **How do we calculate with any number of vectors, at any angles, accurately?**

Graphical methods

Show what is happening; good for sketches and checks; limited accuracy, awkward for many vectors.

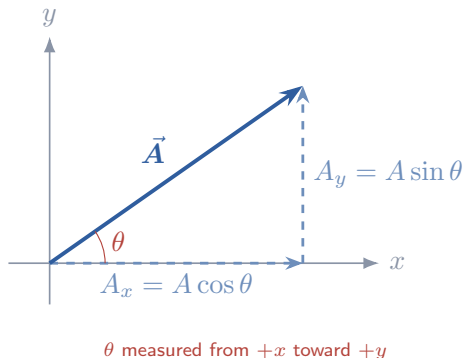
Components

Turn vectors into ordinary numbers; any number of vectors, any accuracy; the same in 3D.

The method of the week

Graphs give geometric understanding; components provide a systematic way to calculate accurately. RESOLVE $\rightarrow$ COMBINE $\rightarrow$ RECONSTRUCT: sketch first, predict, calculate, then check.

But what if... we describe a vector by how much of it lies along each axis? What is that “how much”? $\rightarrow$ **components.**



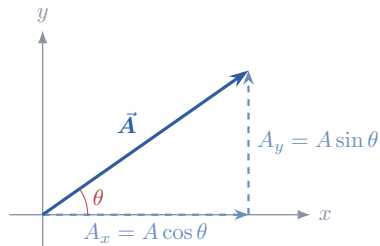
Define. A **component** is the part of a vector along one chosen axis. A_x tells us **how much of $\vec{A}$ lies along the x -direction**; A_y , **how much along the y -direction**. Units: those of the vector. Drawn as a projection on the axis.

Scalar numbers, not vectors

A component such as A_x is a number (positive, negative or zero), as are A_y , A_z . **The corresponding component vector is $A_x \hat{i}$, so**

$$\vec{A} = A_x \hat{i} + A_y \hat{j}.$$

But what if... we know only the magnitude A and the angle θ ? How do we get A_x and A_y ? $\rightarrow$ **resolving.**



θ measured from $+x$ toward $+y$

Rule: resolving (2D)

$$A_x = A \cos \theta, \quad A_y = A \sin \theta$$

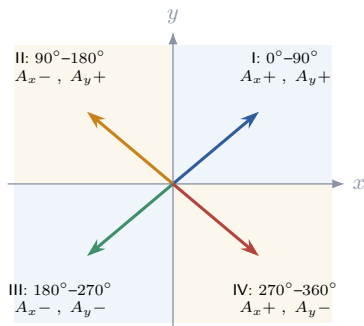
These apply when θ is measured from the $+x$ -axis toward the $+y$ -axis.

Why: $\vec{A}$, A_x , A_y form a right triangle. A_x is *adjacent* to θ , so $\cos \theta = A_x/A$; A_y is *opposite*, so $\sin \theta = A_y/A$.

Predict first

For $\theta = 150^\circ$, are A_x , A_y positive or negative?

But what if... the vector points left or down, as at 150° ? Then $\cos \theta$ or $\sin \theta$ is negative. $\rightarrow$ **signs and quadrants.**



Before using equations, ask: does the vector point **right or left**? Does it point **up or down**?

Sign rule

quadrant	I	II	III	IV
A_x	> 0	< 0	< 0	> 0
A_y	> 0	> 0	< 0	< 0

Permanent rule

Sketch → **predict signs** → **calculate**
The calculator never replaces the sketch.

But what if... the angle is given from another axis, such as 37° from $+y$ or 45° below $+x$? Then $A \cos \theta$ and $A \sin \theta$ swap roles. → **resolving carefully**.

Question

A force of 12.0 N acts at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find F_x and F_y .

Sketch $\rightarrow$ predict signs $\rightarrow$ calculate

Right or left? Up or down? So what signs do you expect?

Question

A force of 12.0 N acts at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find F_x and F_y .

Sketch $\rightarrow$ **predict signs** $\rightarrow$ **calculate**

Right or left? Up or down? So what signs do you expect?

Quadrant II, so $F_x < 0$, $F_y > 0$.

$$F_x = 12.0 \cos 150^\circ = -10.4 \text{ N}$$

$$F_y = 12.0 \sin 150^\circ = +6.00 \text{ N}$$

Evaluate: $\sqrt{10.4^2 + 6.00^2} = 12.0 \checkmark$

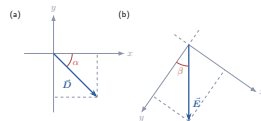
Your turn

Find the x - and y -components of $\vec{D}$ and $\vec{E}$.

(a) $|\vec{D}| = 3.00 \text{ m}$, $\alpha = 45^\circ$ below $+x$.

(b) $|\vec{E}| = 4.50 \text{ m}$, $\beta = 37.0^\circ$ from the $+y$ axis, with tilted axes.

Sketch $\rightarrow$ **predict signs** $\rightarrow$ **calculate**. Young, Example 1.6.



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(a) Counter-clockwise from $+x$, $\theta = -45^\circ$:

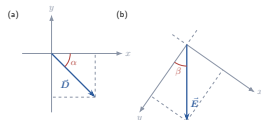
$$D_x = 3.00 \cos(-45^\circ) = +2.12 \text{ m},$$

$$D_y = 3.00 \sin(-45^\circ) = -2.12 \text{ m}$$

(b) β is from $+y$: E_y is *adjacent*, E_x is *opposite*:

$$E_x = 4.50 \sin 37.0^\circ = +2.71 \text{ m},$$

$$E_y = 4.50 \cos 37.0^\circ = +3.59 \text{ m}$$



Your turn

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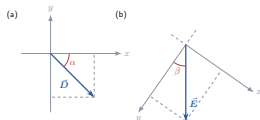
$$D_y = 3.00 \sin(-45^\circ) = -2.12 \text{ m}$$

$A \cos \theta$, $A \sin \theta$ **hold only for θ from $+x$; otherwise ask: adjacent or opposite?**

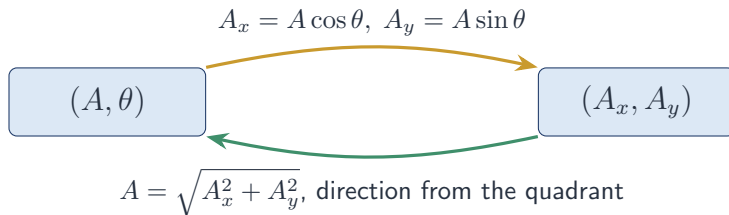
(b) β is from $+y$: E_y is *adjacent*, E_x is *opposite*:

$$E_x = 4.50 \sin 37.0^\circ = +2.71 \text{ m},$$

$$E_y = 4.50 \cos 37.0^\circ = +3.59 \text{ m}$$



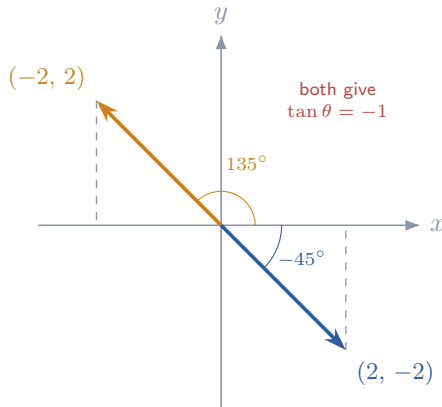
But what if... given A_x and A_y , find A and θ ? A calculator gives the same $\tan^{-1}$ for two vectors that point opposite ways. → **reconstructing**.

**Key idea**

Magnitude/direction and components are two equivalent ways of describing the same vector. Going from components back to magnitude and direction is **reconstructing**.

Predict first

$\vec{P} = (-2, 2)$ and $\vec{Q} = (2, -2)$. Both have $A_y/A_x = -1$, and a calculator gives $\tan^{-1}(-1) = -45^\circ$ for both. Same direction?



Rule: reconstructing a vector

Magnitude: $A = \sqrt{A_x^2 + A_y^2}$.

Quadrant: from the signs of A_x , A_y .

Reference angle:

$$\theta_{\text{ref}} = \tan^{-1} |A_y/A_x|.$$

Direction: I: θ_{ref} ; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

CHECK: the direction agrees with the sketch.

Extension: $\text{atan2}(A_y, A_x)$ builds in both signs; use it after quadrant reasoning. **WATCH OUT:** the same A_y/A_x can mean different directions. **Quadrant first.**

$\vec{P} = (-2, 2)$: left and up, quadrant II.

$$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$$

$$\theta_P = 180^\circ - 45^\circ = 135^\circ$$

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$$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$$

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$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315^\circ} (= -45^\circ)$$

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Your turn: $A_x = -3.0 \text{ m}$, $A_y = -4.0 \text{ m}$. Find A and θ .**But what if...** there are several vectors? How do their components combine? $\rightarrow$ **adding vectors.**

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Your turn: $A_x = -3.0$ m, $A_y = -4.0$ m. Find A and θ . $A = 5.0$ m; quadrant III;

$$\theta_{\text{ref}} = \tan^{-1}(4/3) = 53.1^\circ;$$

$$\theta = 180^\circ + 53.1^\circ = 233.1^\circ \checkmark$$

But what if... there are several vectors? How do their components combine? $\rightarrow$ **adding vectors.**

Question

If $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \dots$, how are the components of $\vec{R}$ related to theirs?

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Rule: adding vectors by components

$$R_x = \sum A_x, \quad R_y = \sum A_y \quad (\text{in 3D also } R_z = \sum A_z)$$

The central skill: RESOLVE $\rightarrow$ COMBINE $\rightarrow$ RECONSTRUCT

RESOLVE each vector (a table helps), **COMBINE** like components, **RECONSTRUCT** the magnitude and direction of $\vec{R}$. **Sketch $\rightarrow$ predict signs $\rightarrow$ calculate.**

CHECK: $\vec{R}$ lies in the quadrant predicted by the sketch, and R is no larger than the sum of the magnitudes.

But what if... the vectors are given as magnitudes and compass directions? $\rightarrow$ **a multiple-vector problem.**

Question

A contestant in a radio race makes three displacements:

$\vec{A}$: 72.4 m, 32.0° **east of north**

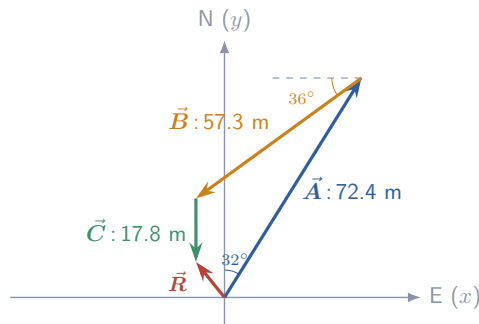
$\vec{B}$: 57.3 m, 36.0° **south of west**

$\vec{C}$: 17.8 m **south**

Find the resultant $\vec{R}$.

Predict first

From the sketch: roughly how far, and in which quadrant?



Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

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	magnitude	θ	x -component	y -component
$\vec{A}$	72.4 m	58.0°	38.37 m	61.40 m
$\vec{B}$	57.3 m	216.0°	-46.36 m	-33.68 m
$\vec{C}$	17.8 m	270.0°	0.00 m	-17.80 m
$\vec{R}$			$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

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$\vec{R}$			$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

Execute (resolve, combine, reconstruct): $R = \sqrt{(-7.99)^2 + 9.92^2} = 12.7 \text{ m}$; left and up, quadrant II; $\theta_{\text{ref}} = \tan^{-1}\left(\frac{9.92}{7.99}\right) = 51.1^\circ$, so $\theta = 180^\circ - 51.1^\circ = 129^\circ$, i.e. 12.7 m at 39° west of north. **Evaluate:** units m; quadrant II, matching the sketch. ✓

Unit vectors $\hat{i}, \hat{j}, \hat{k}$

Direction only: magnitude 1, no units, along $+x, +y, +z$.

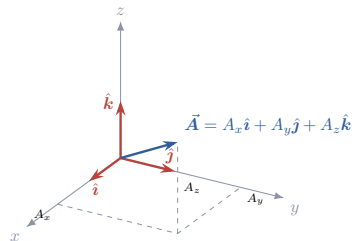
$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

A compact way to write components *together with their directions*.

Number or vector?

A_x, A_y, A_z are **numbers**; $A_x \hat{i}, A_y \hat{j}, A_z \hat{k}$ are **component vectors**.

Direction angles in 3D are extended material (supplement).

**Rule: 3D**

Right-handed axes: curl the fingers from $+x$ to $+y$, the thumb points along $+z$.

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Rule: collect like terms, $\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$.

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

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Your turn

$\vec{A} = (4\hat{i} - 5\hat{j} + 7\hat{k})$ m, $\vec{B} = (\hat{i} - 3\hat{j} + 5\hat{k})$ m, $\vec{C} = (\hat{i} + \hat{j} - 2\hat{k})$ m. Find $\vec{A} + \vec{B} + \vec{C}$ and its magnitude.

But what if... how do we make a unit vector along *any* vector?

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 $(6\hat{i} - 7\hat{j} + 10\hat{k})$ m, $|\cdot| = \sqrt{185} = 13.6$ m

But what if... how do we make a unit vector along *any* vector?

Rule: the unit vector of a vector

$\hat{\vec{A}} = \frac{\vec{A}}{A}$ (divide every component by the magnitude), and conversely $\vec{A} = A \hat{\vec{A}}$. $|\hat{\vec{A}}| = 1$, no units, same direction as $\vec{A}$.

Example 6

$\vec{A} = (3.00\hat{i} - 2.00\hat{j})$ m. Find $\hat{\vec{A}}$ and verify it. **Predict:** are its components bigger or smaller than 1? Which signs, which quadrant?

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IDENTIFY: $A_x = 3.00$ m, $A_y = -2.00$ m. **SET UP:** components of $\hat{\mathbf{A}}$ between -1 and 1 ; $+$ for x , $-$ for y ; quadrant IV.

EXECUTE: $A = \sqrt{3.00^2 + (-2.00)^2}$ m = 3.61 m, $\hat{\mathbf{A}} = \frac{3.00\hat{\mathbf{i}} - 2.00\hat{\mathbf{j}}}{3.61} = 0.832\hat{\mathbf{i}} - 0.555\hat{\mathbf{j}}$

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EVALUATE: $|\hat{\mathbf{A}}| = \sqrt{0.832^2 + 0.555^2} = 1.00$ (exactly $\frac{9}{13} + \frac{4}{13} = 1$); the units cancel; $\tan^{-1}(-2/3) = -33.7^\circ$, quadrant IV, as predicted. ✓

Common mistakes and the correct habit

Mistake**Instead**

Trusting $\tan^{-1}$ for the direction.

Quadrant first from the signs, then the reference angle.

Using $A \cos \theta$, $A \sin \theta$ when θ is not from $+x$.

Ask adjacent or opposite, and take the signs from the sketch.

Adding magnitudes instead of components.

Resolve, combine, reconstruct.

Calculating before predicting the signs.

Sketch, predict the signs, then calculate.

Mixing scalar components with component vectors.

A_x is a number; $A_x \hat{i}$ is the vector.

Calculator in the wrong angle mode.

Check degrees against radians before every trigonometric step.

Lecture 2: motion

Projectile motion: the x - and y -motions are treated separately, using

$$v_x = v \cos \theta \text{ and } v_y = v \sin \theta.$$

Lecture 3: forces

Newton's second law is applied component by component, $\sum F_x = ma_x$ and $\sum F_y = ma_y$, with axes along and perpendicular to a slope.

Beyond mechanics

Electric and magnetic fields and every other vector quantity are added by components in exactly the same way.

Physical meaning

Resolve, combine, reconstruct is a skill you will use every week of this course.

Contact 3: Extend

How do vector products describe physical quantities?

From combining vectors to constructing quantities

So far, we have used vectors to represent and combine physical quantities. We now ask a new question: **what useful physical quantities can we construct from two vectors?**

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Pulling a crate

How much of the pull acts **along the motion?**

pulling a crate $\rightarrow$ **dot product** $\rightarrow$ *work*
 $\text{vector} \cdot \text{vector} \rightarrow$ **scalar**

Turning a door

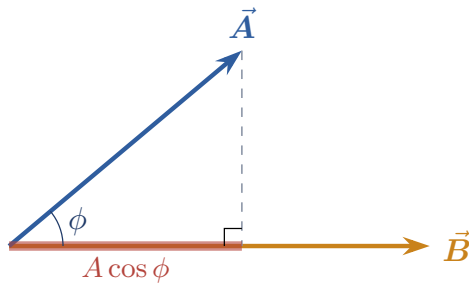
The effect depends on where and which way you push, and has an **axis**.

turning a door $\rightarrow$ **cross product** $\rightarrow$
torque
 $\text{vector} \times \text{vector} \rightarrow$ **vector**

We first understand what each product means geometrically, then learn how to calculate it.

Question

How much of one vector lies along the direction of another?



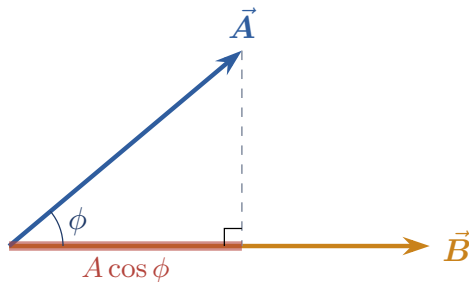
The part of $\vec{A}$ that lies along $\vec{B}$ is the **projection**

$$A \cos \phi$$

where **for dot and cross products**, ϕ is the **angle between the vectors when placed tail-to-tail**, $0^\circ \leq \phi \leq 180^\circ$.

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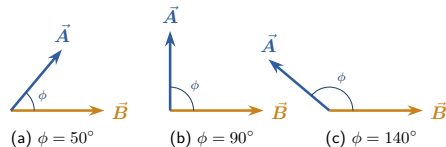
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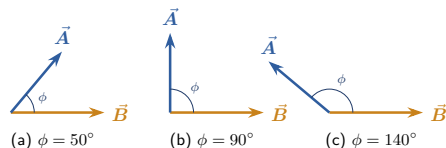
Definition: dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi = (A \cos \phi) B$$

vector $\cdot$ vector $\rightarrow$ **scalar**. Units multiply:
 $\text{m} \cdot \text{m} = \text{m}^2$.

**Predict first**

For each pair: is $\vec{A} \cdot \vec{B}$ positive, zero, or negative?

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Rule: sign of the dot product

$$\phi < 90^\circ \Rightarrow \vec{A} \cdot \vec{B} > 0; \quad \phi = 90^\circ \Rightarrow \vec{A} \cdot \vec{B} = 0; \quad \phi > 90^\circ \Rightarrow \vec{A} \cdot \vec{B} < 0$$

Which angle?

ϕ is the angle between the vectors placed tail to tail. Do not use the internal angle of a head-to-tail triangle: for the cyclist legs of Example 2, $\phi = 60^\circ$ but $\gamma = 120^\circ$.

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

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Rule: unit-vector dot products

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

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Expanding $(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$, only the like terms survive:

Rule: dot product from components

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Connect: work, Lecture 4

$$W = \vec{F} \cdot \vec{s}$$

A constant force $\vec{F}$ acting through a displacement $\vec{s}$. Work is studied fully in Lecture 4.

Question

$|\vec{A}| = 7.50 \text{ m}$ at 53.0° , $|\vec{B}| = 5.00 \text{ m}$ at 130.0° . Find $\vec{A} \cdot \vec{B}$.

Predict first

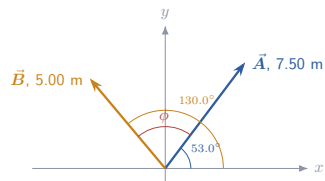
Is ϕ less than or more than 90° ? So what sign?

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after Young, Example 1.9

$$\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$$

$$\vec{A} \cdot \vec{B} = (7.50 \text{ m})(5.00 \text{ m}) \cos 77.0^\circ = 8.44 \text{ m}^2$$

Evaluate: positive since $\phi < 90^\circ$; units are m^2 , not m.

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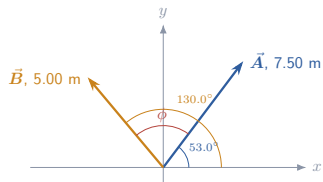
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after Young, Example 1.9

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \cdot \vec{B}$.

Question

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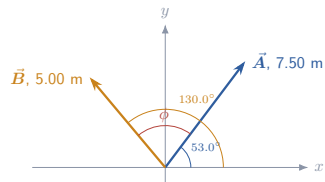
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$$\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$$

$$\vec{A} \cdot \vec{B} = (7.50 \text{ m})(5.00 \text{ m}) \cos 77.0^\circ = 8.44 \text{ m}^2$$

Evaluate: positive since $\phi < 90^\circ$; units are m^2 , not m.



after Young, Example 1.9

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \cdot \vec{B}$.
 $= (1)(-5) + (9)(1) + (2)(-1) = 2$

Question

How can two vectors produce a *third* vector?

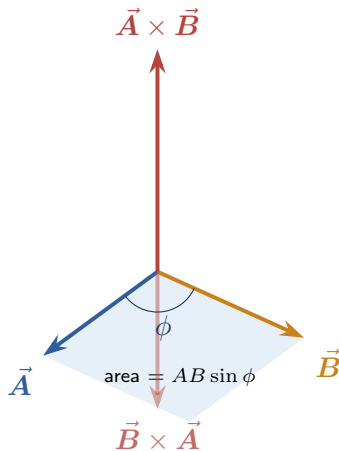
Question

How can two vectors produce a *third* vector?

Definition: cross product

- Magnitude: $|\vec{A} \times \vec{B}| = AB \sin \phi$ (area of the parallelogram)
- Direction: perpendicular to the plane of $\vec{A}$ and $\vec{B}$, given by the **right-hand rule**

The right-hand rule is on the next slide.



Right-hand rule

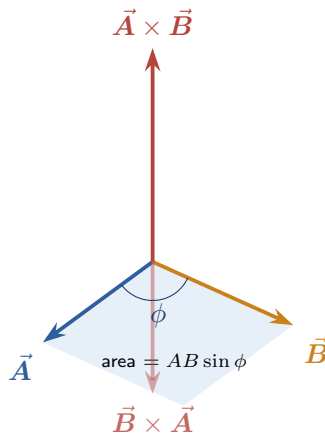
For $\vec{A} \times \vec{B}$:

1. Place $\vec{A}$ and $\vec{B}$ tail to tail.
2. Point the fingers of the right hand along $\vec{A}$.
3. Curl the fingers toward $\vec{B}$ through the smaller angle.
4. The thumb gives the direction of $\vec{A} \times \vec{B}$.

Order matters

$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$. Swapping the vectors reverses the direction.

Special cases: parallel or antiparallel ($\phi = 0^\circ, 180^\circ$): $\vec{A} \times \vec{B} = \vec{0}$, in particular $\vec{A} \times \vec{A} = \vec{0}$. Perpendicular: $|\vec{A} \times \vec{B}| = AB$, the largest possible.



Dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi$$

- Result: a **scalar**
- Zero when the vectors are perpendicular
- Maximum magnitude when the vectors are parallel or antiparallel

Cross product

$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

- Result: a **vector**
- Zero when the vectors are parallel or antiparallel
- Maximum magnitude when the vectors are perpendicular

The angle ϕ in both products

ϕ is the angle between the vectors when placed tail-to-tail, $0^\circ \leq \phi \leq 180^\circ$. **Do not use the internal angle of a head-to-tail vector triangle as ϕ** , such as the $\gamma = 120^\circ$ in the cyclist problem.

Connect: torque, Lecture 8 (CORE)

$\vec{\tau} = \vec{r} \times \vec{F}$, with $|\vec{\tau}| = rF \sin \phi$: a force $\vec{F}$ applied at position $\vec{r}$ from a pivot. The direction comes from the right-hand rule; the order matters; $\tau = 0$ if $\vec{F} \parallel \vec{r}$ and is largest if $\vec{F} \perp \vec{r}$.

REFERENCE: computational extension, cross product from components

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

Unit-vector products

$$\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j} \text{ (cyclic +, reversed -); } \hat{i} \times \hat{i} = \vec{0}.$$

The determinant memory aid and the full equation anatomy are in the Reference deck.

But what if... does the formula give the same vector as the right-hand rule? → **a worked example.**

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Check (components): $\vec{A} = 6\hat{i}$,
 $\vec{B} = 3.46\hat{i} + 2\hat{j}$
 $\Rightarrow (6)(2) - (0)(3.46) = 12$ along $\hat{k}$. ✓

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$ and check it is perpendicular to both.

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Check (components): $\vec{A} = 6\hat{i}$,
 $\vec{B} = 3.46\hat{i} + 2\hat{j}$

$$\Rightarrow (6)(2) - (0)(3.46) = 12 \text{ along } \hat{k}. \checkmark$$

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$ and check it is perpendicular to both.
 $-11\hat{i} - 9\hat{j} + 46\hat{k}$; $(\vec{A} \times \vec{B}) \cdot \vec{A} = -11 - 81 + 92 = 0$, $(\vec{A} \times \vec{B}) \cdot \vec{B} = 55 - 9 - 46 = 0$ ✓

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Check (components): $\vec{A} = 6\hat{i}$,
 $\vec{B} = 3.46\hat{i} + 2\hat{j}$

$$\Rightarrow (6)(2) - (0)(3.46) = 12 \text{ along } \hat{k}. \quad \checkmark$$

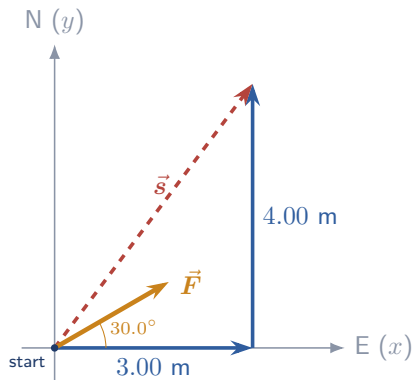
Example 9: integrated application

A worker pulls a crate across a level floor with a rope. The rope force is 50.0 N at 30.0° north of east. The crate moves 3.00 m east, then 4.00 m north.

- (a) Find the crate's displacement $\vec{s}$.
- (b) Find the work done by the rope force, $W = \vec{F} \cdot \vec{s}$.
- (c) The same force acts at $\vec{r} = (0.800\hat{i} + 0.600\hat{j})\text{ m}$ from a pivot. Find the torque $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Sketch first. Which vector tool does each part need, and why? Is $W > 0$? Does $\vec{\tau}$ point along $+z$ or $-z$?



top view; $F = 50.0\text{ N}$; z points out of the page

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$. Predict: $\phi < 90^\circ$, so $W > 0$.

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$. Predict: $\phi < 90^\circ$, so $W > 0$.

Execute.

(a) Add the legs (components):

$$\vec{s} = (3.00\hat{i} + 4.00\hat{j}) \text{ m}$$

$$s = 5.00 \text{ m at } 53.1^\circ \text{ north of east}$$

Resolve the force:

$$F_x = 50.0 \cos 30.0^\circ = 43.3 \text{ N}$$

$$F_y = 50.0 \sin 30.0^\circ = 25.0 \text{ N}$$

(b) Dot product:

$$W = (43.3)(3.00) + (25.0)(4.00) = 230 \text{ J}$$

(c) Cross product; only the $\hat{k}$ term survives:

$$\tau_z = r_x F_y - r_y F_x = (0.800)(25.0) - (0.600)(43.3)$$

$$\vec{\tau} = -5.98 \hat{k} \text{ N m}$$

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$. Predict: $\phi < 90^\circ$, so $W > 0$.

Execute.

(a) Add the legs (components):

$$\vec{s} = (3.00\hat{i} + 4.00\hat{j}) \text{ m}$$

$$s = 5.00 \text{ m at } 53.1^\circ \text{ north of east}$$

Resolve the force:

$$F_x = 50.0 \cos 30.0^\circ = 43.3 \text{ N}$$

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$$W = (43.3)(3.00) + (25.0)(4.00) = 230 \text{ J}$$

(c) Cross product; only the $\hat{k}$ term survives:

$$\tau_z = r_x F_y - r_y F_x = (0.800)(25.0) - (0.600)(43.3)$$

$$\vec{\tau} = -5.98 \hat{k} \text{ N m}$$

Evaluate (verification):

$$Fs \cos \phi = (50.0)(5.00) \cos 23.1^\circ = 230 \text{ J } \checkmark$$

$$rF \sin \phi = (1.00)(50.0) \sin 6.9^\circ = 5.98 \text{ N m } \checkmark$$

Right-hand rule from $\vec{r}$ (36.9°) to $\vec{F}$ (30.0°):

clockwise seen from above, so $-z$ $\checkmark$

$W > 0$, as predicted $\checkmark$

Common mistakes and the correct habit

Mistake**Instead**

Using the internal triangle angle in $\vec{A} \cdot \vec{B}$ or $\vec{A} \times \vec{B}$.

ϕ is the angle tail to tail, $0^\circ \leq \phi \leq 180^\circ$.

Writing a dot product as a vector, or a cross product as a number.

Dot gives a scalar; cross gives a vector.

Forgetting that $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$.

Use the right-hand rule and keep the order.

Swapping cos and sin between the two products.

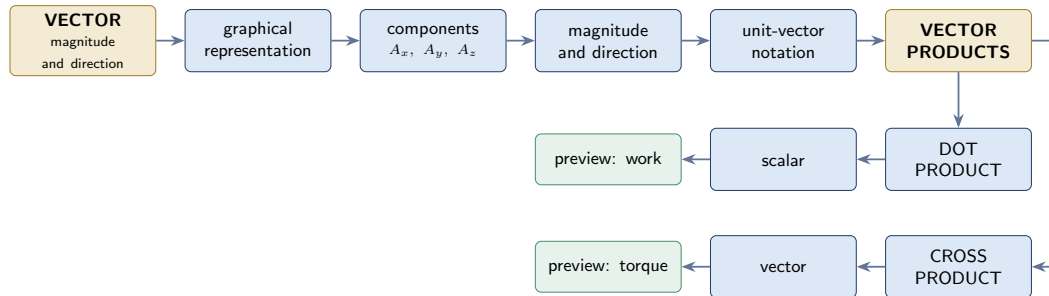
Test with the special cases: perpendicular and parallel.

A wrong sign on the $\hat{j}$ term of the cross product.

Use the determinant, then check $(\vec{A} \times \vec{B}) \cdot \vec{A} = 0$.

Taking $\hat{i} \cdot \hat{j} = 1$.

Like unit vectors give 1, unlike give 0.



Key ideas

- **A vector is a mathematical object used to represent a physical quantity; its components depend on the axes chosen.**
- A_x is a number; $A_x \hat{i}$ is the component vector.
- Magnitudes do not simply add: $|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$.
- **Graphs give geometric understanding; components provide a systematic way to calculate accurately.**
- Magnitude/direction and components describe the same vector.
- **The dot product measures the component of one vector along another; the cross product produces a vector perpendicular to both input vectors.** Their magnitudes are $AB \cos \phi$ and $AB \sin \phi$, respectively.
- Habit: sketch, predict, calculate, evaluate.

Reference tools: where to look

Equations, procedures: the next slides (master toolkit, core equations).

Reference deck: the seven specialised toolkits, equation anatomy, vocabulary, extra examples and practice.

Common mistakes: the last slide of each contact.

MASTER TOOLKIT – For a general vector problem

1. **IDENTIFY** What is being asked?
2. **SKETCH** Draw the physical situation.
3. **CHOOSE AXES** Select useful perpendicular coordinates.
4. **PREDICT** Estimate signs, quadrant, size and direction.
5. **RESOLVE** Convert vectors into components if necessary.
6. **COMBINE** Perform the required vector operation.
7. **RECONSTRUCT** Find magnitude and direction if required.
8. **EVALUATE** Check units, signs, quadrant, magnitude, direction and physical reasonableness.

If the result disagrees with the sketch or prediction, investigate before accepting the answer. The seven specialised toolkits are in the Reference deck.

Inside the course framework

IDENTIFY (step 1): what is being asked?

SET UP (steps 2–4): sketch, choose axes, predict.

EXECUTE (steps 5–7): resolve, combine, reconstruct.

EVALUATE (step 8): check the result.

Idea	Equation	Conditions and units
Along a line	$R = A + B \quad \text{or} \quad R = A - B $	one line only
Perpendicular	$R = \sqrt{A^2 + B^2}, \quad \tan \theta = B/A$	90° between the vectors
Cosine rule	$R^2 = A^2 + B^2 - 2AB \cos \gamma$	γ : internal angle opposite R
Components	$A_x = A \cos \theta, \quad A_y = A \sin \theta$	θ from $+x$; units of $\vec{A}$
Magnitude, direction	$A = \sqrt{A_x^2 + A_y^2}, \quad \theta_{\text{ref}} = \tan^{-1} A_y/A_x $	place by quadrant
Addition	$R_x = \sum A_x, \quad R_y = \sum A_y$	same units for all terms
Unit-vector form	$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$	$\hat{i}, \hat{j}, \hat{k}$ dimensionless
Unit vector of $\vec{A}$	$\hat{A} = \vec{A}/A, \quad \vec{A} = A\hat{A}$	$ \hat{A} = 1$, no units
Dot product	$AB \cos \phi = A_x B_x + A_y B_y + A_z B_z$	scalar; product of units
Cross product	$ \vec{A} \times \vec{B} = AB \sin \phi$	vector; right-hand rule
Cross, components	$(A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$	$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$

- ☐ distinguish a scalar from a vector?
- ☐ add and subtract vectors graphically?
- ☐ resolve a vector with the correct signs?
- ☐ determine the direction of a vector, including its correct quadrant?
- ☐ add several vectors by components?
- ☐ write a vector with $\hat{i}, \hat{j}, \hat{k}$?
- ☐ use IDENTIFY $\rightarrow$ SET UP $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE?
- ☐ calculate a dot product and interpret its sign?
- ☐ calculate a cross product and determine its direction?
- ☐ explain when each product is zero?

Next

Problem sheet 1: extra component, resultant, dot and cross problems, plus challenge problems. The Reference deck holds the supporting material cut from the live lecture; the supplementary notes cover linear independence and direction angles in 3D (extension).